SS1 - Buggy application

VINAYAKA JAGADISH | PES1UG25CS855 | Lab 7

`Command: python bowling_alley_parser.py`

```
==== Running format_player_name ====
Enter a player name for the scoreboard: Vinayaka Jagadish
Scoreboard name: Vinayaka J

==== Running parse_frame ====
Enter a frame as X (strike), 7/ (spare) or two digits (e.g. 45):
Traceback (most recent call last):
  File "/Users/vinayakajagadish/Documents/Codex/2026-11-05/https-github-com-setapesu26-l6-l7/work
/repo/LAB7/.screenshot-run/bowling_alley_parser.py", line 47, in <module>
    main()
    ~~~~^^
  File "/Users/vinayakajagadish/Documents/Codex/2026-11-05/https-github-com-setapesu26-l6-l7/work
/repo/LAB7/.screenshot-run/bowling_alley_parser.py", line 40, in main
    print("Pins (first roll, second roll):", parse_frame(value))
                                              ~~~~~~~~~~~~^^^^^^^
  File "/Users/vinayakajagadish/Documents/Codex/2026-11-05/https-github-com-setapesu26-l6-l7/work
/repo/LAB7/.screenshot-run/bowling_alley_parser.py", line 17, in parse_frame
    first = int(text[0])
                ~~~~^^^
IndexError: string index out of range
(venv) (base) vinayakajagadish@VINAYAKA-MACBOOK-AIR-3 .screenshot-run %
```

An empty frame causes IndexError in the original parser.

SS2 - Tests failing on buggy module

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`Command: pytest test_bowling_alley_parser.py --tb=short`

```
_____________________________ test_parse_scoreboard_no_crash ______________________________
  + Exception Group Traceback (most recent call last):
  |   File "/Users/vinayakajagadish/Documents/Codex/2026-11-05/https-github-com-setapesu26-l6-l7
/work/repo/LAB7/.screenshot-run/test_bowling_alley_parser.py", line 43, in test_parse_scoreboard
_no_crash
  |     @example('')
  |
  |   File "/Users/vinayakajagadish/Documents/Codex/2026-11-05/https-github-com-setapesu26-l6-l7
/work/venv/lib/python3.14/site-packages/hypothesis/core.py", line 1726, in _raise_to_user
  |     raise the_error_hypothesis_found
  | ExceptionGroup: Hypothesis found 2 distinct failures in explicit examples. (2 sub-exceptions
)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "/Users/vinayakajagadish/Documents/Codex/2026-11-05/https-github-com-setapesu26-l6-
l7/work/repo/LAB7/.screenshot-run/test_bowling_alley_parser.py", line 49, in test_parse_scoreboa
rd_no_crash
    |     result = parse_scoreboard(s)
    |   File "/Users/vinayakajagadish/Documents/Codex/2026-11-05/https-github-com-setapesu26-l6-
l7/work/repo/LAB7/.screenshot-run/bowling_alley_parser.py", line 29, in parse_scoreboard
    |     name, score = entry.split(':')
    |     ^^^^^^^^^^^
    | ValueError: not enough values to unpack (expected 2, got 1)
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "/Users/vinayakajagadish/Documents/Codex/2026-11-05/https-github-com-setapesu26-l6-
l7/work/repo/LAB7/.screenshot-run/test_bowling_alley_parser.py", line 51, in test_parse_scoreboa
rd_no_crash
    |     assert False, f'parse_scoreboard crashed: {exc!r}'
    | AssertionError: parse_scoreboard crashed: ValueError('not enough values to unpack (expecte
d 2, got 1)')
    | assert False
    | (note: 1 other explicit example also failed with this error; use Verbosity.verbose to view
)
    | Failing explicit example: test_parse_scoreboard_no_crash(
    |     s='',
    | )
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "/Users/vinayakajagadish/Documents/Codex/2026-11-05/https-github-com-setapesu26-l6-
l7/work/repo/LAB7/.screenshot-run/test_bowling_alley_parser.py", line 53, in test_parse_scoreboa
rd_no_crash
    |     assert all(isinstance(k, str) and k.strip() and type(v) is int
    |                and 0 <= v <= 300 for k, v in result.items())
    | AssertionError: assert False
    |  +  where False = all(<generator object test_parse_scoreboard_no_crash.<locals>.<genexpr>
at 0x107c94310>)
    | Failing explicit example: test_parse_scoreboard_no_crash(
    |     s='Asha:301',
    | )
    +------------------------------------
================================== short test summary info ===================================
FAILED test_bowling_alley_parser.py::test_format_player_name_no_crash - AssertionError: format_p
layer_name crashed: AttributeError("'NoneType' object has no attrib...
FAILED test_bowling_alley_parser.py::test_parse_frame_no_crash - ExceptionGroup: Hypothesis foun
d 3 distinct failures in explicit examples. (3 sub-exceptions)
FAILED test_bowling_alley_parser.py::test_parse_scoreboard_no_crash - ExceptionGroup: Hypothesis
 found 2 distinct failures in explicit examples. (2 sub-exceptions)
================================== 3 failed in 0.16s ==================================
(venv) (base) vinayakajagadish@VINAYAKA-MACBOOK-AIR-3 .screenshot-run %
```

All three completed Hypothesis tests fail on the original module.

SS3 - Fixed application

VINAYAKA JAGADISH | PES1UG25CS855 | Lab 7

`Command: python bowling_alley_parser.py`

```
==== Running format_player_name ====
Enter a player name for the scoreboard: Vinayaka Jagadish
Scoreboard name: Vinayaka J

==== Running parse_frame ====
Enter a frame as X (strike), 7/ (spare) or two digits (e.g. 45): Asha:150,Ravi:132
Pins (first roll, second roll): None

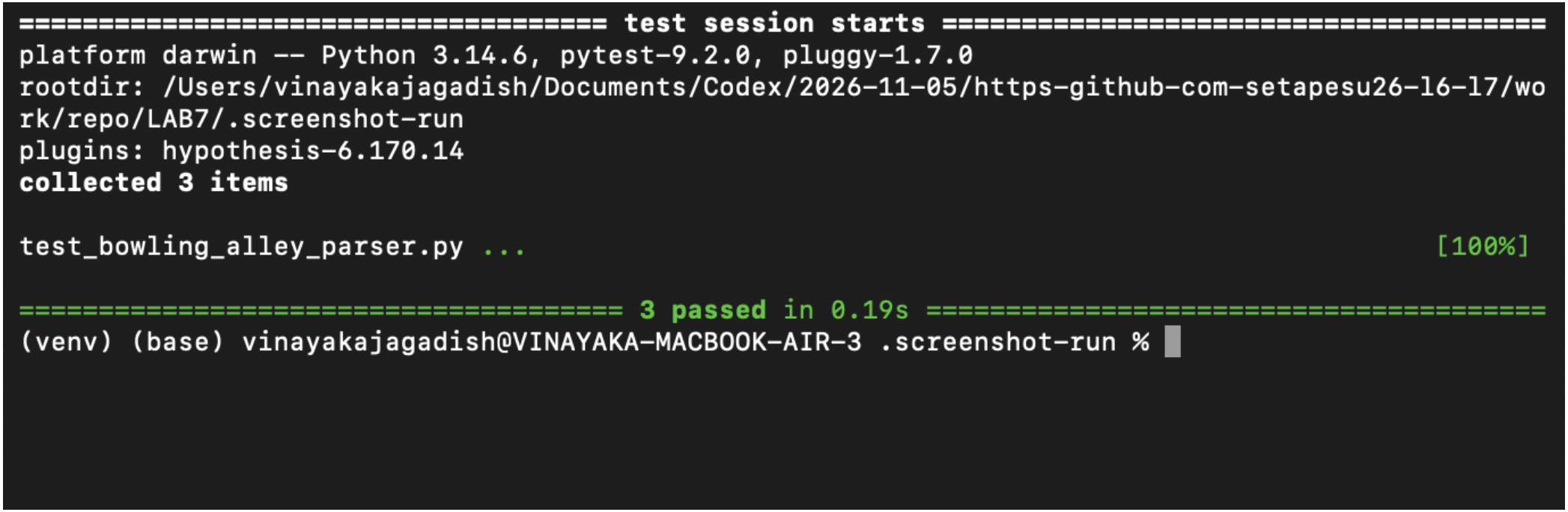
==== Running parse_scoreboard ====
Enter scores as NAME:SCORE separated by commas (e.g. Asha:150,Ravi:132): Asha:150,Ravi:132
Scoreboard: {'Asha': 150, 'Ravi': 132}
(venv) (base) vinayakajagadish@VINAYAKA-MACBOOK-AIR-3 .screenshot-run %
```

Malformed frame input returns None; the valid scoreboard is parsed successfully.

SS4 - Tests passing on fixed module

VINAYAKA JAGADISH | PES1UG25CS855 | Lab 7

`Command: pytest test_bowling_alley_parser.py`



All three Hypothesis tests pass after input validation is added.

Bug analysis and supplemental reflections

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The supplied handout does not contain the three reflection questions mentioned in the repository. The following are supplemental explanations, not answers to unseen questions.

1. What bugs were found? Non-string names cause AttributeError. Empty frames cause IndexError; malformed digits cause ValueError. Frames such as 99 exceed ten pins, and longer strings were silently truncated. Scoreboards crash on missing or extra separators and non-integer scores; empty names, duplicates and out-of-range scores were accepted.

2. How did fuzzing help? Each of the three Hypothesis tests generates Unicode strings, None, numbers, bytes, lists and dictionaries. Explicit regression examples cover valid bowling syntax and invalid boundaries. All three tests failed on the original module and all three passed after the fixes. Passing fuzz tests provides evidence, not proof for every possible Python object.

3. How were the bugs fixed? Validate types before string operations, frame length before indexing, ASCII digits before conversion, and pin totals before returning. Scoreboards validate every pair and scores from 0 to 300. Invalid frames return None; invalid scoreboards return {}; non-string names return an empty string. Scoreboard validation is atomic, and duplicate names are rejected. Leading zeros are removed before integer conversion to avoid the integer-string digit limit.